Belal A. N. Abdelkarim

Mechanical Engineer · R&D Prototyping · 3D Design & Visualization

3d.belal@gmail.com +90 5392727057 Türkiye https://3dbelal.xyz

Summary

Final-year Mechanical Engineering student at Abdullah Gül University, graduating June 2026. Completed two internships at Teknopark startups in Kayseri, designing and delivering real hardware at both. Seven years of professional 3D production in Blender running in parallel — modelling, texturing, animation, rendering. The overlap is deliberate: I design parts, print them, and visualize them.

Education

Abdullah Gül University — Bachelor of Science
Mechanical Engineering — Kayseri, Türkiye • 2022 – 2026

İstanbul Sabahattin Zaim Üniversitesi — Master of Science and Nature
Electrical and Electronics Engineering — 2026-present

Experience

Masswell Advanced Energy Systems — Erciyes Teknopark, Kayseri
Mechanical Design Intern — Aug – Sep 2025

- Designed and 3D printed the full enclosure system for a lithium-ion forklift battery — BMS housing, fire safety holder, power supply container, resistor case, and screen mount
- Built all models in SolidWorks from physical measurements; printed in-house in PLA and iterated on tolerances when parts failed
- Applied FDM anisotropy principles to fix structural column fractures — reoriented print axis and added fillets at stress concentration points, resolving the failure without adding material
- All five components shipped inside a battery delivered to a client

MTM Biotechnology — Erciyes Teknopark, Kayseri
Mechanical Design Intern — Jun – Aug 2025

- Only mechanical designer on three concurrent wearable medical device housings: an EEG headband, an EMG muscle sensor, and a smart earbud
- One project arrived as a verbal description with no sketches or CAD reference — produced a complete 3D model, internal component layout, and photorealistic renders for stakeholder review
- Iterated on enclosure geometry to maintain sensor contact with skin while keeping parts compact and 3D-print-ready; coordinated dimensions with the electronics engineer on each device

Freelance — Remote
3D Artist — 2017 – Present

- Product visualisation, modelling, texturing, and animation in Blender across seven years of client work
- Delivers complete renders from brief to final output — lighting, shading, Cycles rendering, and compositing in DaVinci Resolve and Photoshop

Projects

Self-Balancing Two-Wheel Robot — 2024 – 2025

- Designed the chassis in Blender, 3D printed it, wired the electronics by hand, and wrote the full firmware in Arduino C — the entire system built alone

- Implemented a cascaded PID architecture: an inner tilt-angle loop for balance and an outer velocity-feedback loop for controlled forward and backward movement
- Achieves stable balance within ±0.5° of setpoint using only an MPU6050 IMU and a complementary filter — no wheel encoders, no optical sensors
- Resolved I2C bus corruption from motor EMI through isolated power domains, twisted-pair wiring, and a firmware bus-reset timeout; all PID parameters tunable live over Bluetooth without reflashing

Volunteer

English Speaking Club — Host · Kayseri Gençlik Merkezi 2024 – Present

Kayseri, Türkiye

- Hosted weekly sessions for 8–25 participants over two years; devised topics and created an environment where people felt comfortable making mistakes
- Progressed from member to host

Board Member · AGÜ International Association (AIA) Oct 2024 – Sep 2025

- Organised events for the international student community at Abdullah Gül University

Skills

Mechanical Design

SolidWorks, FDM 3D printing, Blender, DFM, Heat-set inserts

Embedded Systems

Arduino C/C++, Cascaded PID, Sensor fusion, I2C · Bluetooth serial

3D Production

Blender, Modelling, Texturing, Animation, Cycles rendering, DaVinci Resolve

Languages

Arabic (native), English (advanced)